Digital Signal Processing

Laplace and Z Transforms

Prof. Aldebaro Klautau.
Federal University of Pará (UFPA)

www.lasse.ufpa.br & www.ppgee.ufpa.br

Motivation to Laplace transform

Not every signal has representation via Fourier analysis

$$x(t) = e^{3t}u(t) \longrightarrow$$ Integral does not converge **=** No Fourier transform

This can be circumvented by a trick which is <u>pre-multiplying</u> the input signal $x(t)$ <u>by an exponential</u> $e^{-\sigma t}$ obtaining $x(t)e^{-\sigma t}$

Example:

1) Take the signal x(t)

$$x(t) = e^{3t}u(t)$$

2) Use the pre-multiplication trick

$$z(t) = x(t)e^{-4t} = e^{-t}u(t)$$

$$x(t)e^{\boxed{-\sigma}t}$$ this trick!

3) Apply FT <u>on the new signal</u>

$$Z(\sigma,\omega) = \int_{-\infty}^{\infty} z(t)e^{-j\omega t}\mathrm{d}t$$

$$= \int_{0}^{\infty} e^{(-1-j\omega)t}\mathrm{d}t = \frac{1}{1+j\omega}$$

The Laplace transform equations

$$X(s) = \mathcal{L}\{x(t)\} = \mathcal{F}\{x(t)e^{-\sigma t}\} \boxed{= \int_{-\infty}^{\infty} x(t)e^{-\boxed{s}t}\mathrm{d}t}$$

Where $\boxed{s = \sigma + j\omega}$

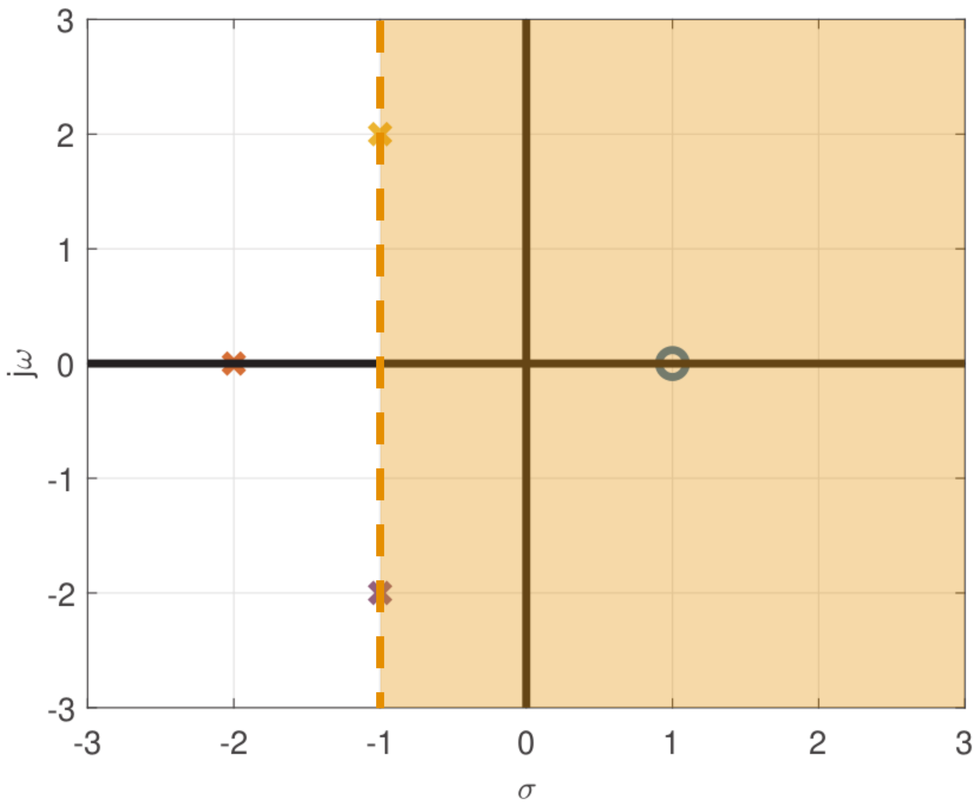
$$\begin{cases} \boxed{X(s) \quad = \quad \int_{-\infty}^{\infty} x(t)e^{-st}\mathrm{d}t \quad \text{Transform}} \\ \boxed{x(t) \quad = \quad \frac{1}{2\pi j}\oint_{\gamma-j\infty}^{\gamma+j\infty} X(s)e^{st}ds \quad \text{Inverse-transform}} \end{cases}$$

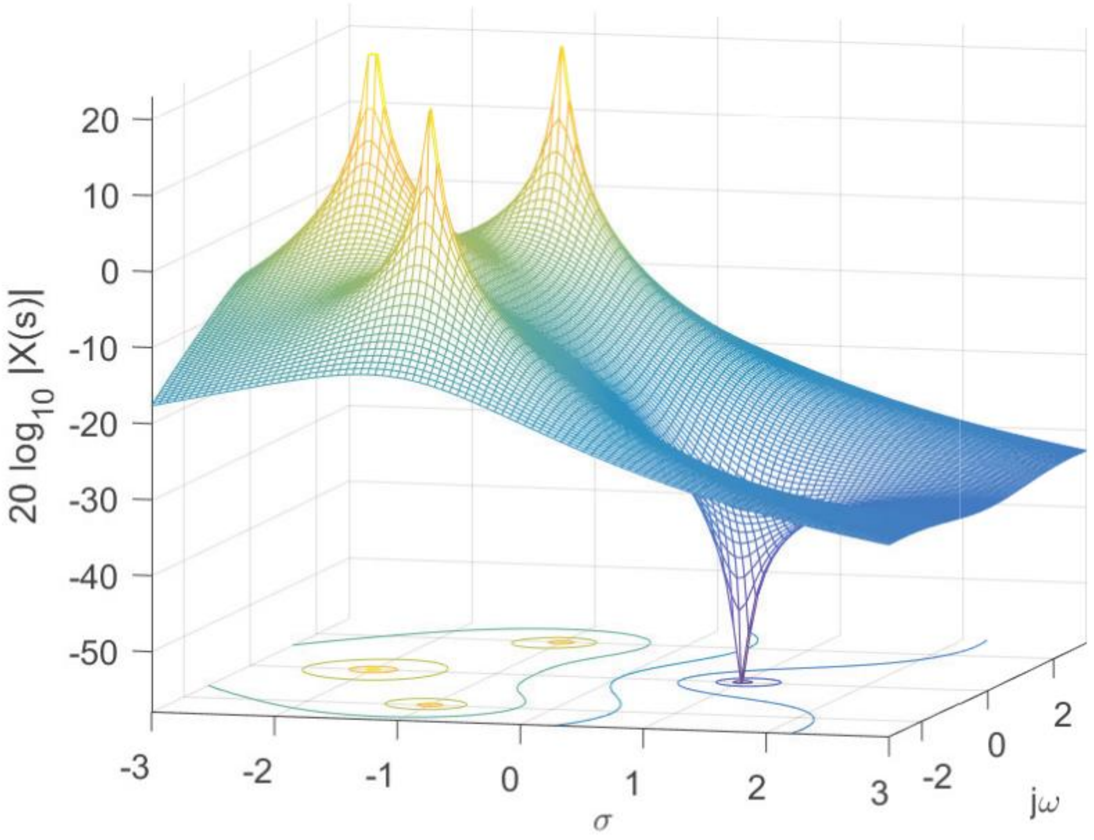
Laplace transform = Fourier transform multiplied by the exponential $e^{-\sigma t}$

Zeros and poles

$$X(s) = \frac{s-1}{s^3+4s^2+9s+10} = \frac{s-1}{(s+2)(s+1-j2)(s+1+j2)}$$

ROC $\sigma > -1$

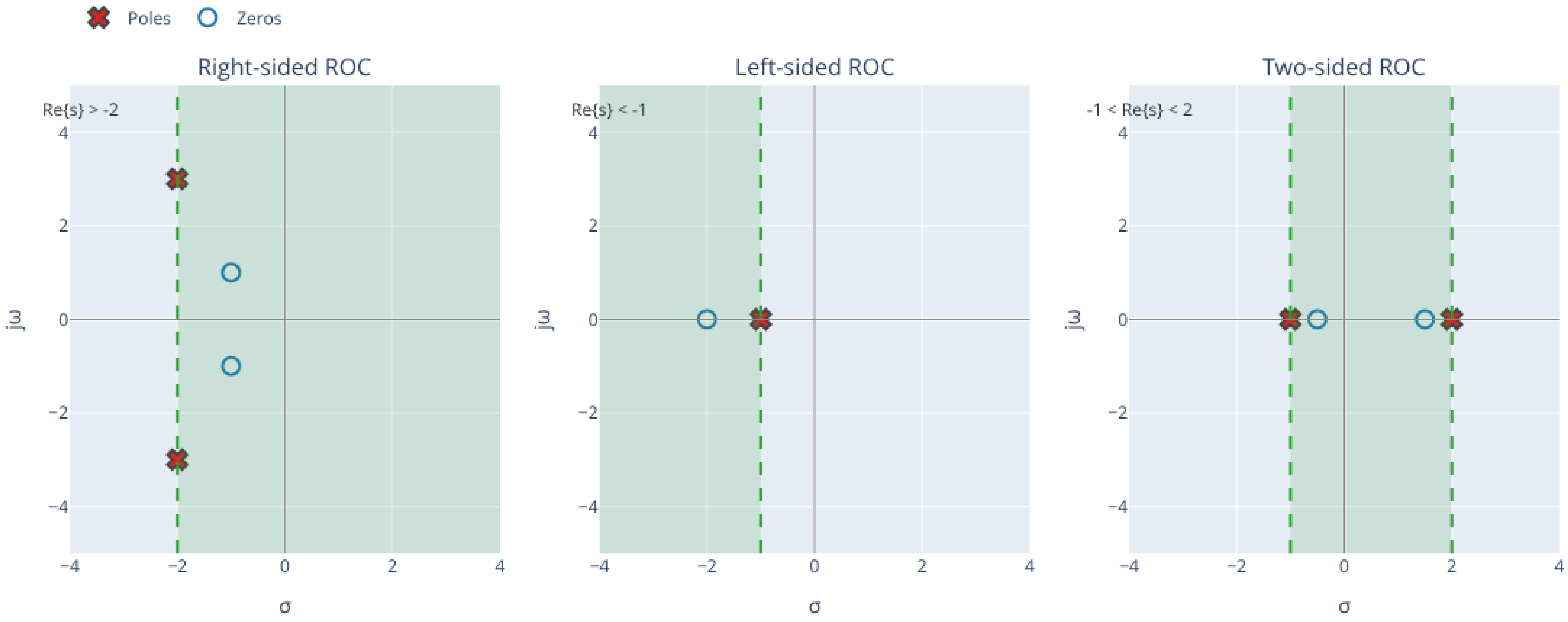




Region of Convergence (ROC)

$$X_1(s) = \frac{(s+1)^2+1}{(s+2)^2+9} \qquad X_2(s) = \frac{s+2}{s+1} \qquad X_3(s) = \frac{(s+0.5)(s-1.5)}{(s+1)(s-2)}$$

The ROC of $\mathcal{L}\{x(t)\}$ is a connected region in the s-plane for which the integral converges



***Pay attention*: For the original x(t) to have a FT, ROC must include the jω axis**

Z transform

Is the counterpart of the Laplace transform for discrete-time signals

$$\begin{cases} X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n} & \text{Transform} \\ x[n] = \frac{1}{2\pi j} \oint_C X(z) z^{n-1} dz & \text{Inverse-transform} \end{cases}$$

C is a counterclockwise closed path encircling the origin and is entirely in the region of convergence (ROC)

The contour (or path) C must encircle all poles of X(z)

ROC

$$X(z) = \frac{z + 0.9}{(z - 0.8)(z - 0.5 - j0.6)(z - 0.5 + j0.6)}$$

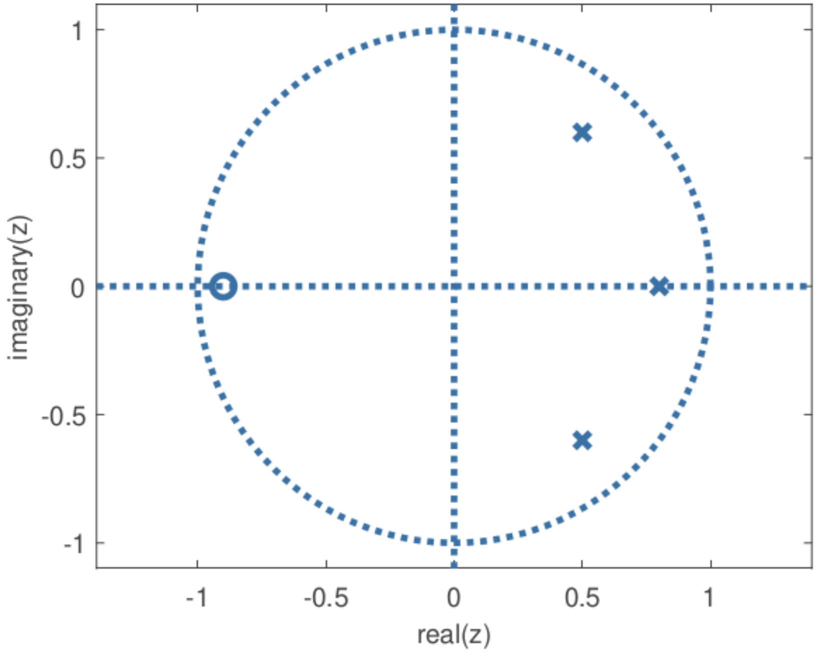
ROC cannot be inferred from the expression alone, you still need information from x[n].

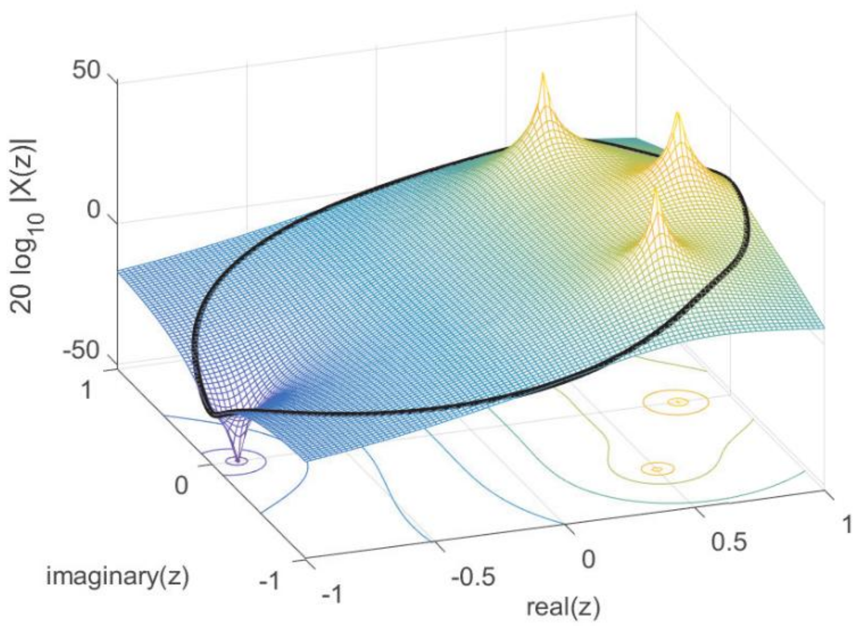
Possible ROCs

$|z| < \sqrt{0.61}$

$\sqrt{0.61} < |z| < 0.8$

$|z| > 0.8$





Pay attention: For the original x[n] to have a DTFT, ROC must include the unit circle

Some Z transform Properties

- $\delta[n] \Leftrightarrow 1$
- $x[n-n_0] \Leftrightarrow X(z)z^{-n_0}$

$$\boxed{\delta[n-n_0] \Leftrightarrow z^{-n_0}}$$

Example 1: By inspection, we calculate the Z-transform of

$$g[n] = 3\delta[n+4] + 2\delta[n] - 2.5\delta[n-3],$$

as:

$$G(z) = 3z^4 + 2 - 2.5z^{-3}$$

Do not forget the ROC!

Terms after $n = 0$ need $|z| \neq 0$

Terms before $n = 0$ need $|z| \neq \infty$

ROC: $0 < |z| < \infty$

Z transform of an exponential

In cases x[n] is a left or right-sided exponential, we can use the geometric series

$$\sum_{n=0}^{\infty} \alpha r^n = \frac{\alpha}{1-r}, |r| < 1$$

Example 2: the Z transform of $x[n] = a^n u[n]$ is obtained as follows

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n} = \sum_{n=-\infty}^{\infty} a^n u[n] z^{-n} = \sum_{n=0}^{\infty} a^n z^{-n} = \sum_{n=0}^{\infty} (a/z)^n$$

$$\boxed{\begin{array}{l} \alpha = 1 \\ r = az^{-1} \\ |a/z| < 1 \end{array}}$$

$$a^n u[n] \Leftrightarrow \frac{1}{1 - az^{-1}} = \frac{z}{z-a}, |z| > |a|$$

The same expression is used for a left-sided exponential $x[n] = -a^n u[-n-1]$ with $|z| < |a|$

Inverse Z-transform via partial fractions

$$X(z) = \frac{8z - 19}{(z-2)(z-3)}$$

$x[n]$ is right-sided

$$X(z) = \frac{A}{(z-2)} + \frac{B}{(z-3)}$$

$$Az + Bz = 8z$$
$$-3A - 2B = -19$$

$$A = 3, B = 5$$

$$X(z) = \frac{3}{(z-2)} + \frac{5}{(z-3)}$$

We want to force the the appearance of terms like

$$\frac{z}{z-a}$$

$Y(z) = zX(z)$

$y[n-1] = x[n]$

$$Y(z) = \frac{3z}{(z-2)} + \frac{5z}{(z-3)}$$

$$y[n] = [3(2^n) + 5(3^n)]u[n]$$

$$x[n] = [3(2^{n-1}) + 5(3^{n-1})]u[n-1]$$

Thank you!

Prof. Aldebaro Klautau.
Federal University of Pará (UFPA)

www.lasse.ufpa.br & www.ppgee.ufpa.br